Chapter 5, Section 5

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1. Let f be a rational function on the surface X. Show that it is possible to "resolve the singularities of f" in the following sense: there is a birational moprhism $g : X' \to X$ so that f induces a morphism of X' to $\mathbb{P}^1$.

2. Let $Y \cong \mathbb{P}^1$ be a curve in a surface X, with $Y^2 < 0$. Show that Y is contractible (5.7.2) to a point on a projective variety X_0 (in general singular).

3. If $\pi : \tilde{X} \to X$ is a monoidal transformation with center P, show that $H^1(\tilde{X}, \Omega_{\tilde{X}}) \cong H^1(X, \Omega_X) \oplus k$. This gives another proof of (5.8).

4. Let $f : X \to X'$ be a birational morphism of nonsingular surfaces.

(a) If $Y \subseteq X$ is an irreducible curve such that $f(Y)$ is a point, then $Y \cong \mathbb{P}^1$ and $Y^2 < 0$.

(b) (Mumford [6].) Let $P' \in X'$ be a fundamental point of f^{-1}, and let $Y_1, \ldots, Y_r$ be the irreducible components of $f^{-1}(P')$. Show that the matrix $\|Y_i.Y_j\|$ is negative definite.

5. Let C be a curve, and let $\pi : X \to C$ and $\pi' : X' \to C$ be two geometrically ruled surfaces over C. Show that there is a finite sequence of elementary transformations (5.7.1) which transform X into X'.

6. Let X be a surface with function field K. Show that every valuation ring R of K/k is one of the three kinds described in (II, Ex 4.12).

7. Let Y be an irreducible curve on a surface X, and suppose there is a morphism $f : X \to X_0$ to a projective variety X_0 of dimension 2, such that $f(Y)$ is a point P and $f^{-1}(P) = Y$. Then show that $Y^2 < 0$.

8. *A surface singularity.* Let k be an algebraically closed field, and let X be the surface in $\mathbb{A}^3_k$ defined by the equation $x^2 + y^3 + z^5 = 0$. It has an isolated singularity at the origin $P = (0, 0, 0)$.

(a) Show that the affine ring $A = k[x, y, z]/(x^2 + y^3 + z^5)$ of X is a unique factorization domain, as follows. Let $t = z^{-1}$, $u = t^3x$, and $v = t^2y$. Show that z is irreducible in A, $t \in k[u, v]$, and $A[z^{-1}] = k[u, v, t^{-1}]$. Conclude that A is a UFD.

(b) Show that the singularity at P can be resolved by eight successive blowing-up. If $\tilde{X}$ is the resulting nonsingular surface, then the inverse image of P is a union of eight projective lines, which intersect each other according to the Dynkin diagram $\mathbb{E}_8$:

Here each circle denotes a line, and two circles are joined by a line segment whenever the corresponding lines intersect.